BSH/W-24/CS/AI/IT/DS30001/03.

Education to Eternity

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

B. Tech ~~Summer~~ / Winter – 2024

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: CSE/AI/IT/DS **Sem:- III**

Subject: NMDM/NTDM /DMSA **Subject Code: CS/AI/IT/DS3M001**

Date of Exam : 21/11/2024 Marks:100 **Time:3hrs**

Instructions:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to: (Listed Below CO & Mention it on question)

CO1: Describe the concept of Numerical methods, Bisection method and Newton-Raphson Method and their convergence, Gauss Elimination method and Crout's method.

CO2: Illustrate the concept of mathematical expectation, variance and standard Deviation, moment and moment generating function, characteristic function, conditional expectation, Variance and moment.

CO3: Apply the concept of joint probability and random variable, probability function Of CRV, joint probability function of DRV, joint distribution functions of DRV.

CO4: Analyse the problem by using the concept of linear dependence and independence of vector, Consistency of homogeneous and non-homogeneous equation.

CO5: Evaluate the problem based on the concept of Eigen values and Eigen vectors, Sylvester's theorem, Quadratic form, Canonical form, differentiation & integration of matrices.

CO6: Students will be able to create the new concept by using the theory of Group, group, Subgroups, Homomorphism, cosets, Lagrange's theorem and Normal subgroup.

SECTION A

Q. No.	Solve the two words or two sentences (All Questions are compulsory).	Bloom's Level & CO	Marks
Q.1	Define: Conditional Probability function of X given Y	L1/CO1	2
Q.2	Recall: Canonical Form.	L1/CO1	2
Q.3	Explain why (Z, ×) does not forms a group ?	L2/CO2	2
Q.4	Illustrate the term: Independent Variables	L2/CO2	2
Q.5	Tell the formula for Newton-Raphson Method.	L1/CO1	2
Q.6	Interpret the term: Algebraic Structure.	L2/CO2	2
Q.7	Outline which of the following sets are group under operation addition R, N, Q & Z.	L2/CO2	2

Q.8	Illustrate the term: Semi-group.	L2/CO2	2
Q.9	Let G be a finite group of order 12 then what are the possible orders of its subgroups?	L1/CO1	2
Q.10	Define: E(X) for DRV & CRV	L1/CO1	2
Q.11	Illustrate the term: Standard Deviation.	L2/CO2	2
Q.12	Define : Characteristic Equation.	L1/CO1	2

SECTION B

Q. No.	**Solve the following Descriptive Type Questions** **(All Questions are compulsory).**	**Bloom's Level & CO**	**Marks**
Q.13	List the three types of Moments and define it for both DRV & CRV	L4/CO4	4
Q.14	Identify the order of each element of the given group G = {0,1,2,3} with addition modulo 4.	L3/CO3	4
Q.15	Solve the matrix to find the eigen values : $\begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$	L3/CO3	4
Q.16	Let X be a random variable with probability distribution as X: 1, 2, 3 P(x): 0.3, 0.2, 0.5 Solve: E(X) & Var(X)	L4/CO4	4
Q.17	The matrix is reduced to the diagonal form $B = \begin{bmatrix} 6 & 0 & 0 \\ 0 & \frac{7}{3} & 0 \\ 0 & 0 & \frac{16}{7} \end{bmatrix}$ Inspect the following terms: 1) Canonical Form 2) Rank 3) Index 4) Signature	L4/CO4	4
Q.18	Solve and find the roots of the function $f(x) = 3x - cosx - 1 = 0$ in the interval [0.6, 0.7] up to two iterations.	L3/CO3	4

SECTION C

Q. No.	**Solve the following Descriptive Type Questions** **(Solve any 04 out of 06).**	**Bloom's Level & CO**	**Marks**
Q.19	Let X be a random variable having probability function $f(x) = \begin{cases} c(x - x^3) & ,0 \le X \le 1 \\ 0 & ,otherwise \end{cases}$ Identify the two moments about origin.	L3/CO3	7
Q.20	Test for the consistency of the non-homogeneous system of equations $5x + 3y + 7z = 4$ $3x + 26y + 2z = 9$ $7x + 2y + 10z = 5$	L4/CO4	7
Q.21	Apply the Newton-Raphson Method to find the root of the equation $xsinx + cosx = 0$ with $x_0 = 3.1416$ by up to forth iterations.	L3/CO3	7

Q.22	Simplify the matrix to find the Eigen vectors for the respective eigen values 2, 3, 6. $$\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$	**L4/CO4**	**7**
Q.23	Evaluate by Bisection Method to find the roots of the function $x^4 - x^3 - x^2 - 4$ in the interval [1,9] up to 4th iteration.	**L5/CO5**	**7**
Q.24	Solve the system of equations by using Gauss- Seidel Method up to two iterations. $$2x - 3y + 20z = 25$$ $$20x + y - 2z = 17$$ $$3x + 20y - z = -18$$	**L6/CO6**	**7**

SECTION D

Q. No.	**Solve the following Descriptive Type Questions** **(Solve any 02 out of 03).**	**Bloom's Level & CO**	**Marks**
Q.25	Let X and Y be random variable having joint probability function $$f(x) = \begin{cases} cxy & , x = 1,2,3\,; y = 1,2,3 \\ 0 & , \quad otherwise \end{cases}$$ Evaluate the following: 1) C 2) P(X>1) 3) Marginal probability function of X & Y 4) P(1 ≤ X ≤ 2 , Y=3) 5) P(X=3, Y=2) 6) Check whether X and Y are independent.	**L5/CO5**	**12**
Q.26	Prove that the set { 1,2,3,4,5,6} of order 6 is finite abelian group under the multiplication modulo 7 as composition.	**L5/CO5**	**12**
Q.27	Solve the system of equations by using Crout's Method $$4x + y - z = 13$$ $$3x + 5y + 2z = 21$$ $$2x + y + 6z = 14$$	**L6/CO6**	**12**